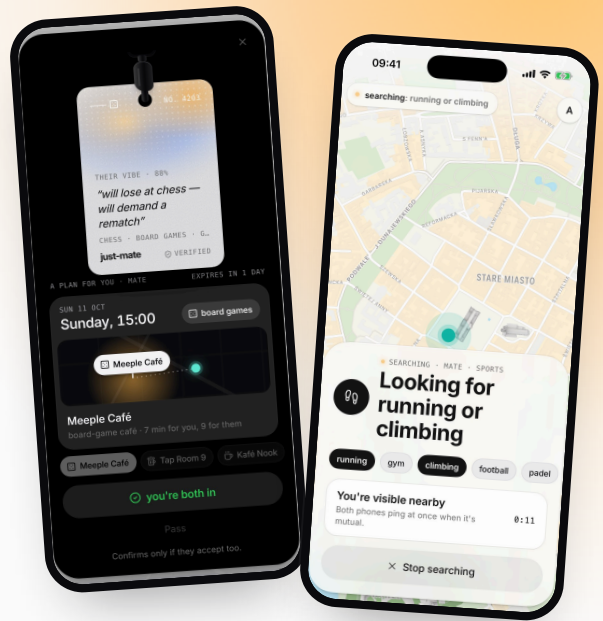




Meet for real.

A plan near you, a person who fits, and someone waiting when you get there.

PRODUCT & TECHNICAL WHITEPAPER · OCTOBER 2026



ABSTRACT

Loneliness affects one in six people worldwide and is linked to more than 871,000 deaths a year¹. For a lonely person, the hard part is rarely finding someone online. It is the step out of the door: organising anything feels like too much, nobody may come, and photo-first apps make meeting feel like an audition.

JustMate takes those steps off them. From a faceless profile, the app proposes one concrete 1:1 plan: a compatible person, an activity both picked, an open public venue within a 15-minute walk of both, and a time both are free. The plan is on only when both accept. A compass then guides them to each other with a bearing and a distance bucket, never a position, and first names unlock only when they meet. For people already out, Now pings two compatible people nearby who want the same thing at the same moment.

This paper describes what the build does today, and labels everything else **PRODUCTION PATH**. JustMate is not therapy and makes no health claims.

1:1

plans, on only when both accept

15 min

max walk to the venue, for each of you

18

public venues in Kraków, every sport covered

0

photos, chats or location history

TEAM

Arthur Kozubov · Jozef Zvalo · Andrej Zak · Nikita Orlov · Serhii Vielkin

CODE & DOCS

github.com/uteg-labs/just-mate
MIT licence

CONTENTS

Problem · Who it's for · How it works · Matching · Safety & privacy · Health case · Business · Built vs production path · References

Loneliness and inactivity are health risks at population scale.

1 in 6

people worldwide are affected by loneliness¹

871,000 2×

deaths a year are linked to loneliness, about 100 every hour¹

lonely people are twice as likely to get depressed¹

31%

of adults, 1.8 billion people, don't get enough physical activity⁵

17–21%

of 13–29-year-olds are lonely, the highest of any age group¹

13%

of people in the EU feel lonely most or all of the time²

+50%

higher likelihood of survival with stronger social relationships⁴

Apps solved matching. The friction is the step out of the door: **no reason to go, no one sure to be there, and a fear of being judged.**

BARRIER 1

Initiative

Low energy makes organising anything, even choosing between options, feel like too much. "Find people now" assumes you are already out and already brave.

BARRIER 2

The empty table

"What if nobody comes?" One no-show confirms the story that nobody wants you.

BARRIER 3

Judgment

Photo-first apps turn meeting into an audition, and chat turns it into weeks of performance before anything real happens.

DESIGN CONSEQUENCE

Every mechanic in JustMate removes one of the three barriers. And because friendship takes about 50 hours together to go from acquaintance to casual friend³, the product's job is not the first meeting but the second and the tenth. Existing apps earn from attention spent on the phone; JustMate succeeds when the phone goes back in the pocket.

Young adults new to a city, with no one to call on a weekday evening.

Primary: 18–35-year-olds who moved for work or study, remote workers, students who haven't found their people, people after a breakup. They can walk, and cities give the density plans need. Loneliness peaks at this age: 17–21% of 13–29-year-olds are lonely¹. Secondary: people already out alone, like a traveller on an evening walk, which is the Now loop.

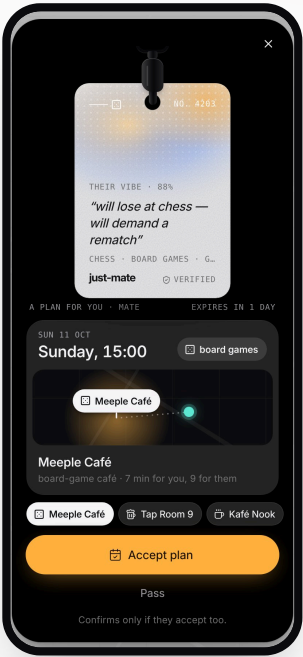
PERSONA · Tomek, 28

Moved to Kraków for a job. Works from home. Hasn't spoken to anyone all weekend.

Not "depressed", just flat and stuck. Organising anything feels impossible; a Meetup room of 40 strangers feels worse. Swipe apps gave him chats that died.

Sunday night a plan drops in: **board games · Sunday 15:00 · Meeple Café**, 7 minutes on foot for him and 9 for them, with a vibe line instead of a face: *"will lose at chess — will demand a rematch"*. He taps **Accept plan**. When the other side accepts too, both phones show **you're both in**.

At 14:45 the compass opens and walks him there. At the table: **"Say hi to Ola."** and how far he walked. Then **Same again next week?** He didn't plan anything, didn't chat with anyone, and knew someone would be there.



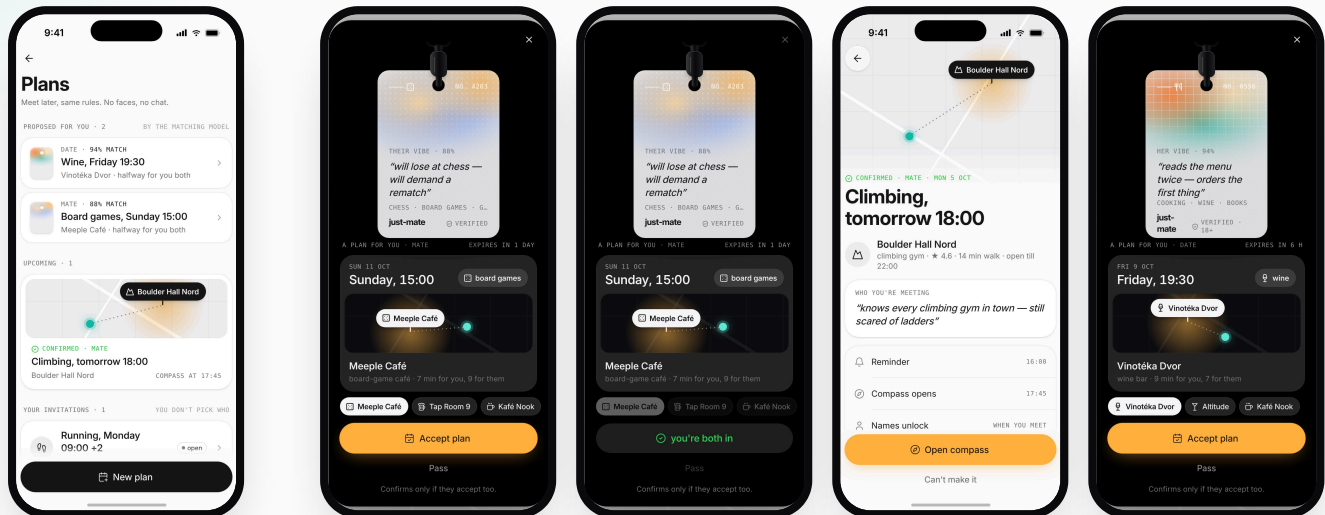
A plan for you
their vibe, the place, the time,
minutes on foot for each

THE ARC WE DESIGN FOR

couch	yes	door	table	again
flat, no energy to plan	one tap: Accept plan	the compass opens 15 min before	an activity to do side by side	"Same again next week?"
WHO	SITUATION	LOOP	WHAT JUSTMATE DOES	
Newcomer	Lonely, low energy, never initiates	Plan	Proposes one concrete plan; he only taps Accept plan	
Shy person	A date feels like an interview	Plan	An activity first: board games, climbing, a run, at a public venue	
Has the energy today	Knows what and where, not who	Plan for later	Their own invitation, offered to one compatible person at a time	
Already out	Alone on an evening walk	Now	Pings a compatible person nearby who wants the same thing	

The app makes the plan. You only say yes.

Plans are strictly 1:1, in both Mate and Date. They need no chat: the card carries the person's vibe, the activity, the place, the time and when the compass opens.



Plans

proposed for you, upcoming, your invitations

A plan for you

alternatives as chips

you're both in

on both phones at once

Confirmed

compass 15 min before

A Date plan

same rules, 18+ only

- Proposed from real profiles and free times.** The matcher pairs two compatible people and picks an activity both chose, a time inside both people's free slots, and an open public venue within a 15-minute walk of both, the one where nobody walks much further. The card shows walking minutes for each: "7 min for you, 9 for them".
- Accept, pass, or suggest a place.** **Accept plan** confirms only if both accept; "waiting for them..." turns into **you're both in** on both phones at the same moment. Picking an alternative venue turns the button into **Suggest this place**. **Pass** is silent: the other side only sees the proposal expire.
- Confirmed.** The plan card shows the venue, who you're meeting as a vibe, and when the compass opens. Names still unlock only when you meet. **Can't make it** is one tap; the other side sees "plan cancelled", no reason asked.
- The walk.** The compass opens 15 minutes before the start and runs for up to 30 minutes, exactly like Now: a bearing and a distance bucket, never a pin. At the table: **We met**.

PLACES

18 public venues in Kraków, hand-tagged with activities and opening hours, with every sport covered. Plans only ever happen at public venues, never at an address.

COPY RULES

Nobody is told "X declined" or "X passed". A passed proposal simply expires. Declines are silent by construction.

NOT IN THE BUILD

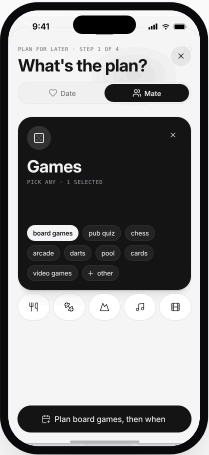
Group plans, status chips, a leave-at time and push reminders are **PRODUCTION PATH**. Without push, plans update while the app is open.

Your own plan, or someone nearby right now.

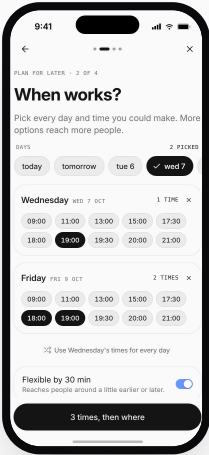
Plan for later: what, when, where, send

YOU DON'T PICK WHO

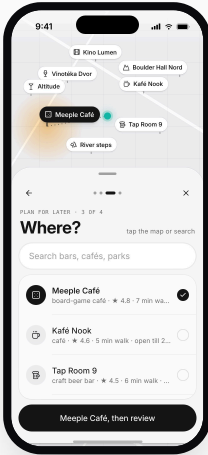
For the days you have the energy. Pick an activity, several days and times (optionally flexible by 30 minutes) and a public venue. The invitation is offered to **one compatible person at a time**. The first one in picks one of your times; you see their vibe and **Confirm plan** or **Pass**, and then it goes to someone else, who only sees "plan filled".



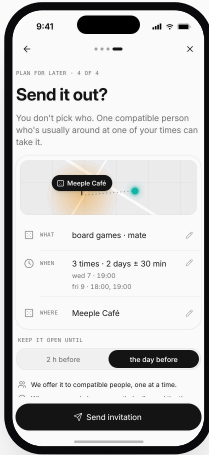
1 · What



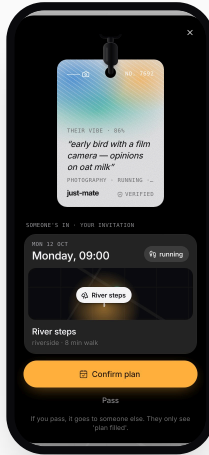
2 · When



3 · Where



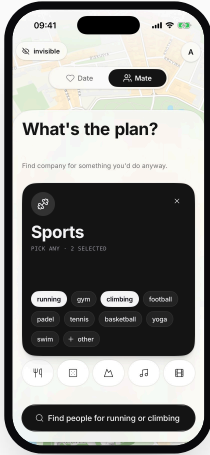
4 · Send



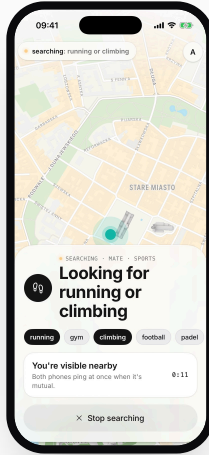
Someone's in

Now: already out

- 01 Pick what you're up for, a category and one or more things inside it, and tap **Find people**. You are invisible until then.
- 02 Both phones ping at once when a compatible person nearby wants the same thing. The default radius is a 10-minute walk.
- 03 Mutual accept. The compass unlocks only if both accept. The session lasts 10 minutes.
- 04 Vanish at any moment: one tap, and the session ends for both.



Pick
sports, games, food...



Searching
visible only now

The compass: distance as a game, deliberately imprecise

SERVER SENDS BEARING + BUCKET

cold over 200 m	warm under 200 m	hot under 80 m	burning under 30 m
--------------------	---------------------	-------------------	-----------------------

An arrow turns with the phone's own heading; colour and haptics escalate as you get closer. The partner's coordinates never reach your phone.

SETTING	VALUE
Walk up to 5 / 10 / 15 min	400 / 800 / 1,200 m
Default Now radius	800 m
Now compass session	10 min
Plan compass opens · runs	T-15 min · 30 min

AFTER MEETING

"Say hi to Ola."

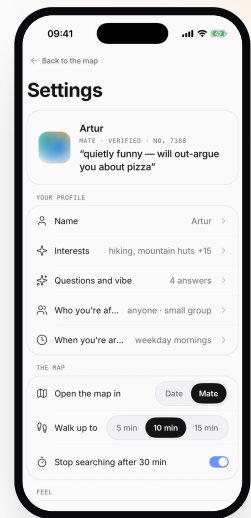
Only the first name unlocks, with the distance you walked to get there. **Same again next week?** is a button; scheduling the pair again is **PRODUCTION PATH**. **Report** is here and on the compass (page 7).

A faceless profile, an explainable score, a learned model.

Onboarding in about two minutes

- 01 **Mate or Date.** Friends and training partners, or someone to fall for. Switchable on the map.
- 02 **Interests.** At least three; each pick opens related ones.
- 03 **Four questions** written by gpt-4o-mini from your picks, each with four tap-able answers. Fixed questions if the model is unavailable.
- 04 **A one-line vibe** in two lowercase clauses. Reroll until it fits. It is all a match ever sees.
- 05 **One selfie**, described once as text, then dropped: never stored, never shown. The check is simulated in this build.

Every saved profile passes OpenAI moderation; a flagged person is never matched.



Your vibe in Settings
no photo, ever

Hard gates: never learned, never scored

- Same mode, and at least one shared activity: the context of the match
- Within walking distance: the shorter of both walk settings (Now), or a venue within 15 minutes of both (Plan)
- Adults are never matched with minors; Date is 18+ only
- Gender and age preferences accepted both ways; not blocked, not paused

The explainable score

$$\begin{aligned} \text{compat} &= 0.7 \times \text{Jaccard}(\text{interests}) \\ &+ 0.3 \times \text{shared intent} \\ \text{match} &\Leftrightarrow \text{gates pass} \wedge \text{compat} \geq 0.45 \end{aligned}$$

Example: 3 shared of 7 distinct interests and a shared intent
→ $0.7 \times 0.43 + 0.3 = 0.60$, a match.

The learned model: a Siamese network for Now

Two texts	Embeddings	Shared encoder	Match head	ONNX over HTTP
who I am (self) and who I'm looking for (target)	OpenAI text-embedding-3-small	projects both into one compatibility space	scores A's target against B's self, both ways	a scorer container the server calls

WHERE	WHAT SCORES THE PAIR	WHY
Now, real accounts	The Siamese model	Falls back to the explainable score when no model score exists
Plans	The explainable score	Shown as a match percentage, defensible line by line
Stage demo	The explainable score	Deterministic for Tomek and Ola

Honest limit. Before launch there are no real meetings to learn from, so the model is trained on synthetic profiles with rule-based labels. We make no quality claims from those labels. Retraining on real outcomes (who met, who said same again) is **PRODUCTION PATH**.

Meeting strangers is the risk. Safety is the architecture.

The first question about any proximity app is "isn't this a stalker tool?" JustMate is built so the answer is no by construction. Every safety feature is free, forever.

CONSENT

Mutual by construction

Both must accept before anything unlocks: a plan, a compass. A pass is never forwarded.

LOCATION

A bearing, never a position

The server relays a direction and a distance bucket. A partner's lat/lng never reaches a phone; the protocol has no field for it.

HISTORY

No location history

Positions live in server memory only while you search or walk, and are dropped when the session ends. Nothing to leak.

EXIT

Vanish

One tap, no confirmation: the session ends for both, instantly, and looks the same whichever side pressed it.

CONTENT

Moderation, no chat

No messaging means no DM channel. Every saved profile passes OpenAI moderation; flagged people are never matched.

AGE

Age gates

Adults are never matched with minors, enforced on the server. Date requires 18+.

REPORT

Report → block

From the compass or after meeting. It blocks that person for you and ends the session.

ABUSE

Two reports → pause

Two independent reports pause the reported person from matching and plans.

HELP

Support lines

Settings carries "Need to talk to someone?" with 116 123 for adults and 116 111 for children and teens.

PRODUCTION PATH

WHAT IT ADDS

Phone and ID verification

One account per number; an optional ID check. Today the selfie check is simulated.

Women-only plans

A free filter, with verified-only matching

Meeting point first

The first compass leads both to a public spot, not to each other

Trusted contact

Share the plan's venue and time through the share sheet

Human review

A person reviews a paused account before any ban; an appeal path

STORED

Account email and password hash; profile: mode, first name, age, interests, answers, vibe, preferences; plans: who and which venue, never where they are.

NEVER

Any photo of you. Location history. Messages. Mood or health data. Anyone's coordinates on your phone.

GDPR / RODO: location is processed only inside an explicit, session-scoped choice to search or walk, and never stored. A data protection impact assessment before launch is **PRODUCTION PATH**. The demo runs on test data only.

Meet, move, look up. And come back again.

JustMate changes what people do, not how they feel about it: they meet someone in person, walk to get there, and put the phone away while they talk. The value comes from repeated, active, in-person contact.

16 of 44

Mate activities are sport or outdoors

5–15 min

on foot before every meeting

0

feeds, likes or streaks to keep you on the phone

PATHWAY	WHAT JUSTMATE DOES	WHY IT MATTERS
Meet	Every flow ends face to face. No chat to get stuck in; "Same again next week?" points at the second meeting.	Stronger social relationships: 50% higher likelihood of survival ⁴ . Friendship takes ~50 hours together ³ .
Move	Every meeting starts on foot: a 5-, 10- or 15-minute walk. 16 of 44 Mate activities are sport or outdoors (running, climbing, padel, football, yoga, cycling...), each with a public venue in Kraków.	31% of adults don't get enough physical activity ⁵ . A partner turns "I should" into a fixed time.
Look up	No feed, no likes, no streaks: nothing to scroll. Time in the app is an anti-metric.	Time in company with the phone away.
Small step	A vibe instead of a face, a mutual yes, a fixed time and a public place. Declines are silent.	Going out becomes possible on a low-energy day.

THE LINE WE HOLD

Not therapy. A reason to go out.

No health claims. Software that treats a condition is a medical device under EU MDR 2017/745. We never say JustMate treats anything.

No mood data. Mood and health are special-category data under GDPR Art. 9. We never ask how you feel.

No guilt mechanics. No streaks to lose, no "you haven't gone out in a week".

Signposting. Support lines in Settings. The app never plays counsellor.

How we will measure it, without health data

PLANNED

SHOW-UPS

Share of confirmed plans where both people tap We met. Target: at least 80% of confirmed attendees.

REPEAT MEETINGS

Share of active users who meet the same person at least twice in 30 days. Repetition is how a friendship starts.

LONELINESS · UCLA-3

The 3-item UCLA Loneliness Scale at sign-up and after 8 weeks: opt-in, anonymous, aggregated, never stored on a profile.

Also watched: the share of meetings that are sport or outdoors, and minutes walked to meet, in aggregate only. Anti-metric: time in the app should not grow.

[3] Hall, 2018 · [4] Holt-Lunstad et al., 2010 · [5] WHO, 26 Jun 2024 · activity count from the CATEGORIES list in @justmate/protocol

Venues pay. People meet for free.

Meeting is never paywalled and safety is never paywalled. Those who earn from people going out, venues and events, pay most. Every figure on this page is **our estimate**, to be replaced with real quotes.

STREAM	HOW IT WORKS	PRICE (EST.)
Partner venues PRIMARY	Cafés, board-game bars, climbing gyms list plan slots for off-peak tables. They see counts and check-ins, never identities.	~€39 / month
Prepaid first round	An optional €5 voucher at partner venues, sold as a wallet top-up; doubles as a commitment device.	15% kept
JustMate+	Reach and convenience, not access: several open invitations at once, travel mode, private plans.	€4.99 / month
Events and festivals	Official plans for a known, opt-in crowd.	€500–2,000

Back of the envelope: one city, 5,000 monthly active users

MONTHLY COST	ASSUMPTION	€	MONTHLY REVENUE	ASSUMPTION	€
Hosting	API, PostgreSQL, backups	~100	Venues	25 × €39	~975
LLM	~1,000 new users × ~€0.01	~20	Prepaid round	900 vouchers, net of card fees	~380
Phone checks	1,000 × ~€0.07	~70	JustMate+	150 × €4.99, net of VAT and fees	~520
ID checks	40% of new users × ~\$0.80	~300	Events	1 festival a quarter	~330
Moderation	20 h × €15	~300	Total		~2,200
Stores, email	Apple fee, email plan	~30			
Total		~820			

Readout: about 21 partner venues at ~€39 cover a city's ~€820 running cost on their own; everything else is margin toward a team.
Source: our estimates in docs/PRODUCT.md §16.3.

REJECTED ON PURPOSE

"Five free connections, then pay". It charges lonely people for the one thing they came for and rewards rationing meetings. Never: ads in the meeting flow, selling data, a paid safety feature.

LAUNCH: PLANS LOWER THE DENSITY BAR

Now needs two people in the same minute; a plan needs one compatible person within a walk over a day or two. A city opens with plans, one campus and a few partner venues first.

Built and demoed, or labelled production path.

BUILT AND DEMOED

- **Plans, 1:1:** proposed from real profiles and free times, Accept plan, you're both in, Suggest this place, silent Pass, Plan for later one person at a time
- **Now:** both phones pinged at once, mutual accept, 10-minute session
- **Compass:** bearing + bucket relay, haptics; Vanish
- **After meeting:** first name and distance walked
- **Safety:** Report blocks and ends the session; two reports pause; moderation; age gates; support lines
- **Matching:** explainable score; Siamese model for Now on real accounts
- **Onboarding:** LLM questions and vibe line
- 18 public venues in Kraków · English, Polish, Slovak · 134+ server tests in CI · demo mode with Tomek and Ola

PRODUCTION PATH

- "Same again next week?" scheduling (today a button)
- Phone and ID verification (selfie check simulated today)
- Women-only plans, meeting point first, trusted contact, human review
- Group plans, status chips, push reminders
- Partner venues and prepaid first round
- UCLA-3 outcome survey with a research partner
- Retraining the model on real meetings; DPIA

STACK

Expo dev-client app (iOS, Android) · Bun + Elysia + Drizzle on PostgreSQL · one WebSocket per phone · a typed protocol package shared by app and server · OpenAI gpt-4o-mini and moderation · ONNX scorer over HTTP

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- [3] Hall, J. A. (2018). How many hours does it take to make a friend? Journal of Social and Personal Relationships. <https://doi.org/10.1177/0265407518761225>
- [4] Holt-Lunstad, J., Smith, T. B., Layton, J. B. (2010). Social Relationships and Mortality Risk: A Meta-analytic Review. PLoS Medicine. <https://doi.org/10.1371/journal.pmed.1000316>
- [5] WHO, Nearly 1.8 billion adults at risk of disease from not doing enough physical activity, news release, 26 June 2024. <https://www.who.int/news/item/26-06-2024-nearly-1.8-billion-adults-at-risk-of-disease-from-not-doing-enough-physical-activity>

Figures from the repository, not external: 16 of 44 Mate activities, 18 venues, 134+ tests. Business figures are our estimates.

AI use: AI tools helped with ideation and boilerplate; the idea, mechanic and solution are the team's. In the product, gpt-4o-mini writes onboarding questions and vibe lines, and OpenAI moderation screens profiles. A Siamese model trained on synthetic profiles scores Now matches; plans use an explainable score. Maps: Apple Maps and Google Maps. Type: Inter.

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